

Short analysis – Seasonality of Wind Speeds in Switzerland

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Context. This short document presents insights about *potential* electricity production in the wind turbines sector in Switzerland. I use Meteoswiss data² and a code I previously developed³.

In short, I gather wind related data at station level across the country. These data are particularly granular, since measures are available at 10-minute intervals from 1950⁴ to today (October 7, 2026). I display visual outputs showing mean wind speed throughout year. The corresponding code is available at this [address](#).

Findings. Overall, it is confirmed that mean wind speed is above its annual mean on the 2015-2025 period *during winter months* (usually December until May). I check it at the country level and at the canton level. Of course there is variation between cantons but on average it is indeed verified. Therefore it suggests that wind turbine electricity production is higher during winter months (December to May), since wind blows faster. I find a similar result as [Cauz et al. \[2025\]](#) who wrote “This distribution between solar PV production (mainly occurring in the summer) and wind production (predominantly during the winter) persists over multiple years”

Graphics. First I produce a graph at the country level (1). On average, between 2015 and 2025, mean wind speed is higher than the annual mean from December to May⁵. I performed the same analysis across all the cantons using a heatmap where dark green represents monthly mean wind speeds above the annual mean (2). In appendix I have displayed results canton by canton (AnnexeA).

1. HEG — Genève, HES-SO

2. Available [here](#).

3. Available on my GitHub page [here](#).

4. Or later, depending on the stations and measurement technologies available at the time.

5. Yearly mean wind speed is at 2.60 m/s. We can note that on average a wind turbine requires at least a wind of about 3 m/s to 5 m/s to start to produce electricity.

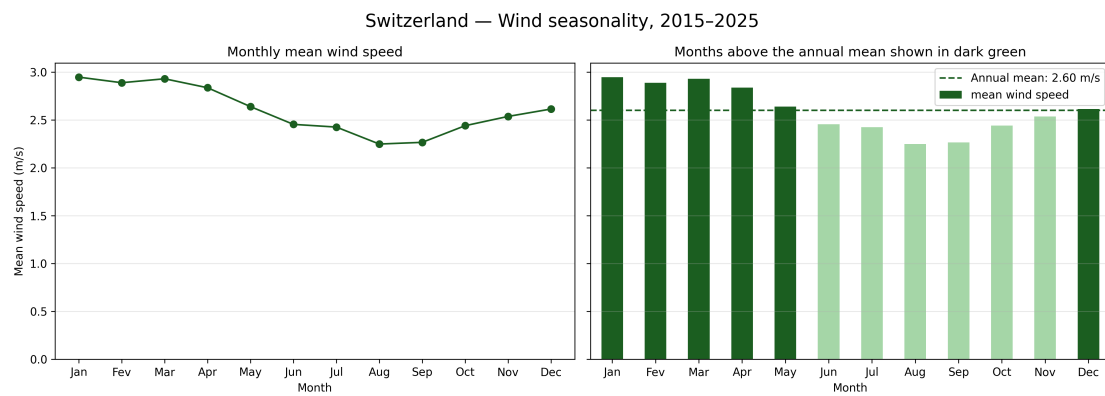


FIGURE 1 – Switzerland seasonality

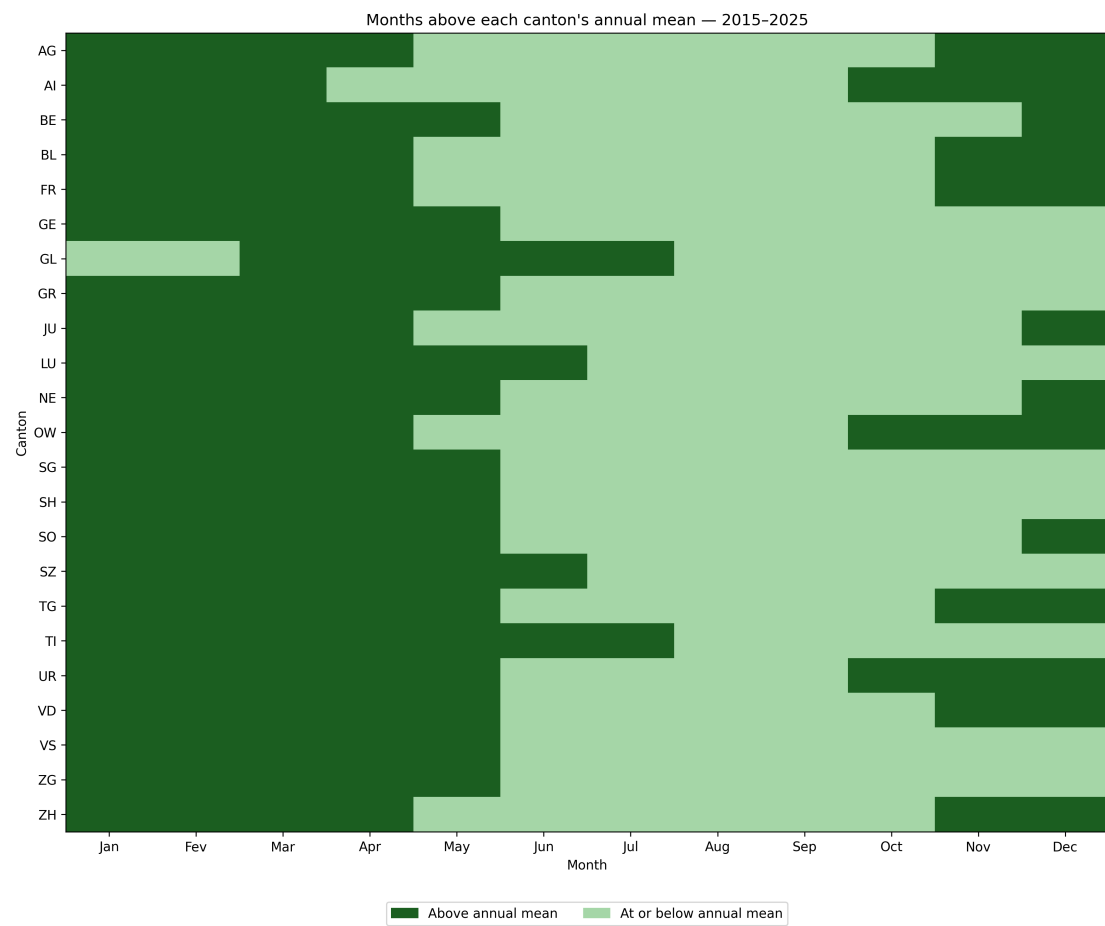
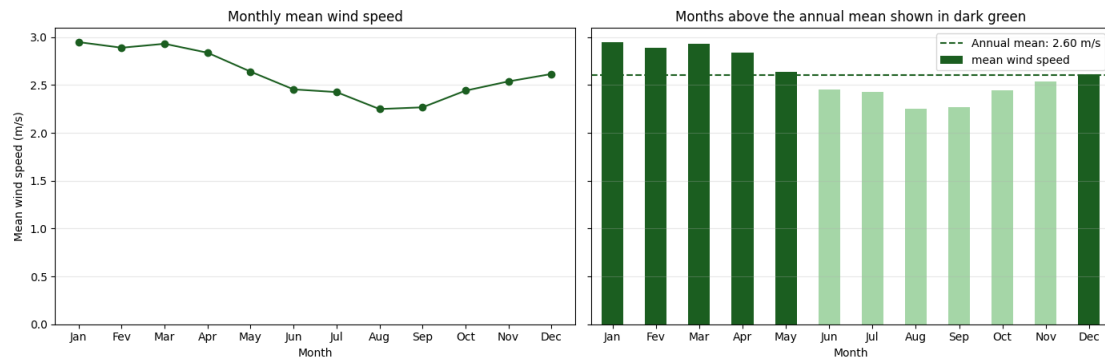


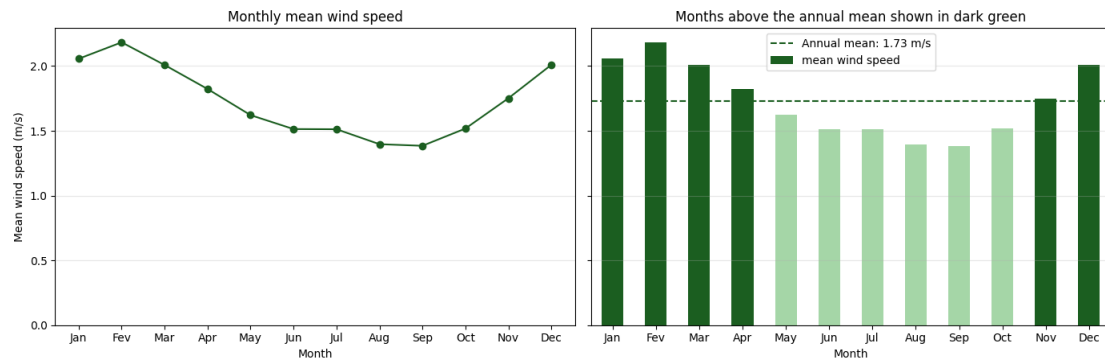
FIGURE 2 – Seasonality by canton

A Results for each canton

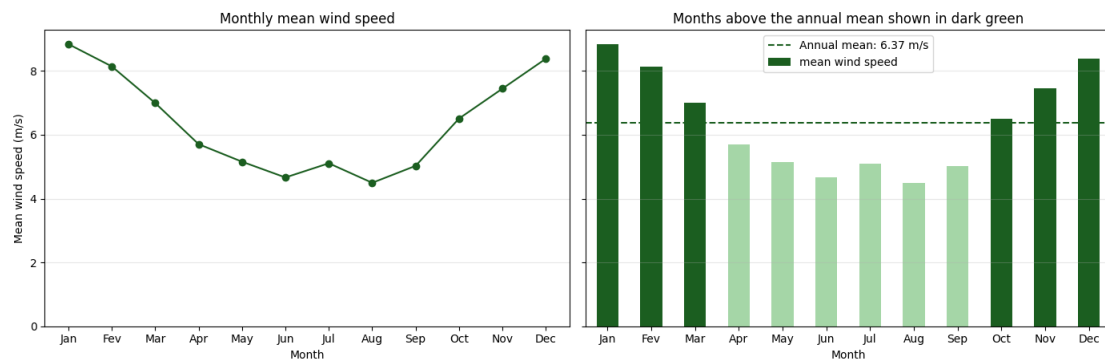
Switzerland — Wind seasonality, 2015-2025



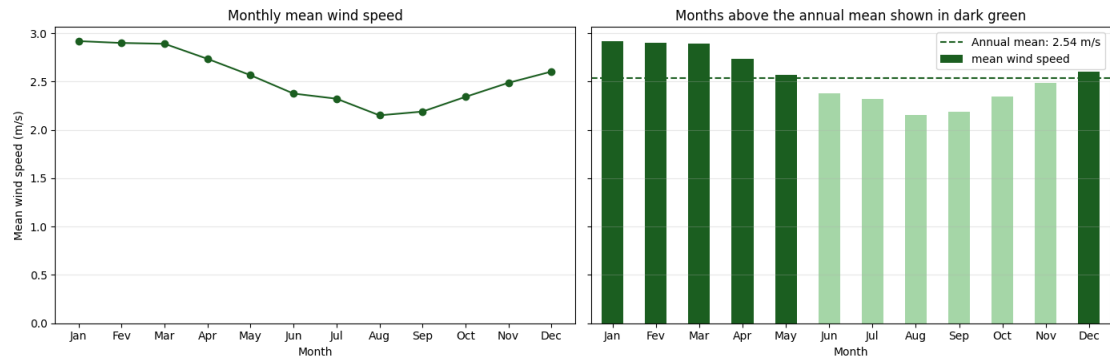
Canton AG — Wind seasonality, 2015-2025



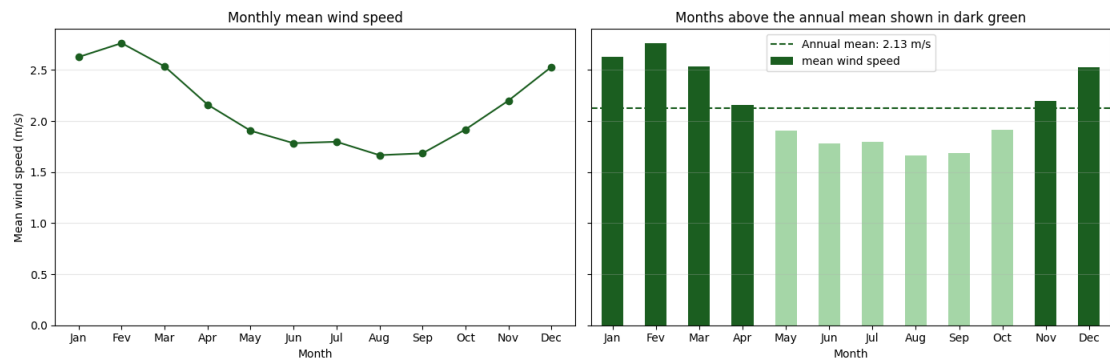
Canton AI — Wind seasonality, 2015-2025



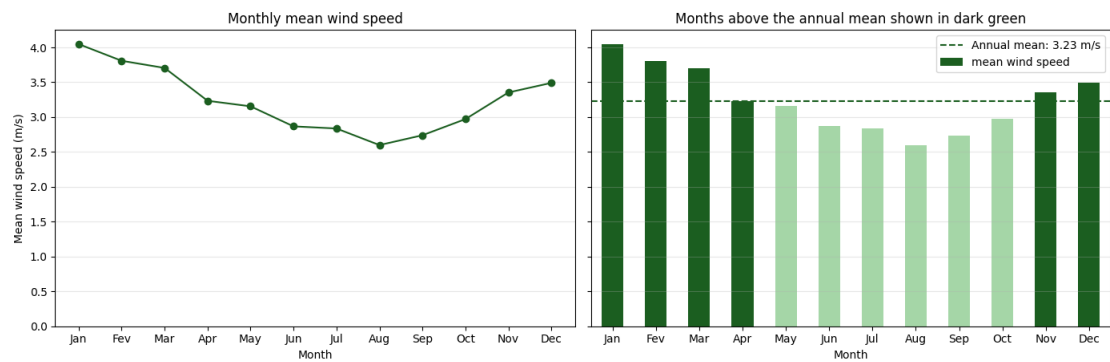
Canton BE — Wind seasonality, 2015-2025



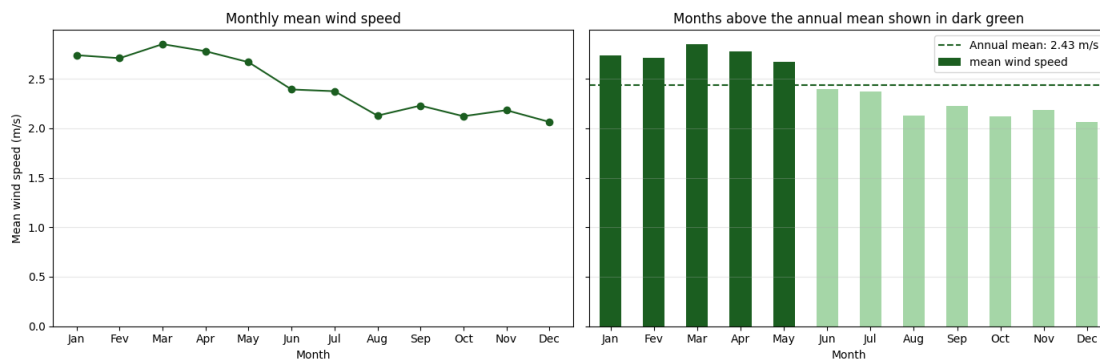
Canton BL — Wind seasonality, 2015-2025



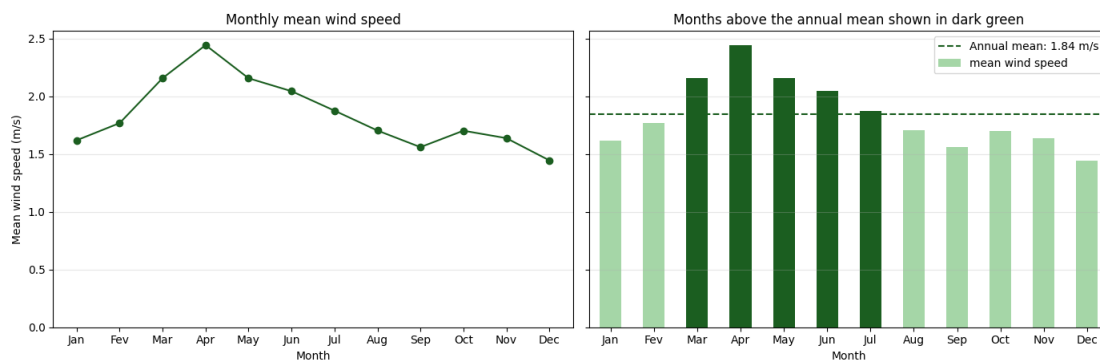
Canton FR — Wind seasonality, 2015-2025



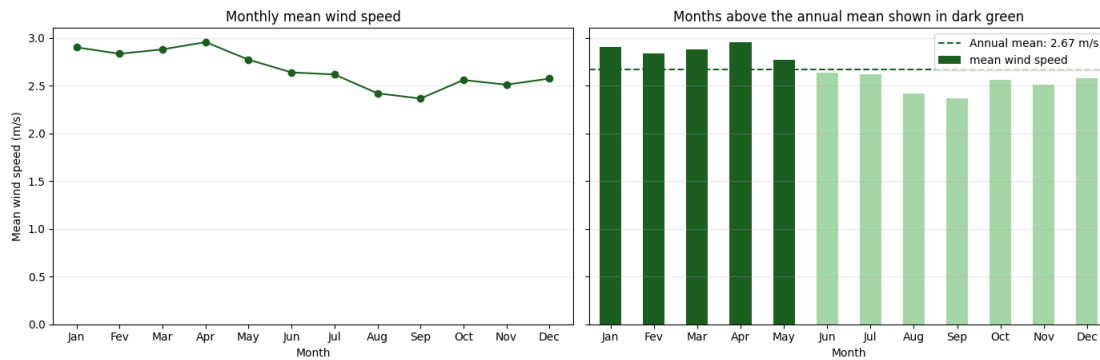
Canton GE — Wind seasonality, 2015–2025



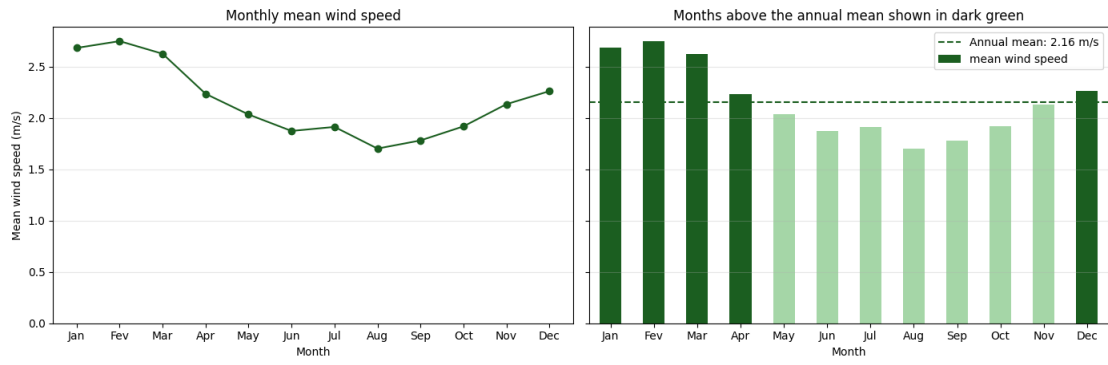
Canton GL — Wind seasonality, 2015–2025



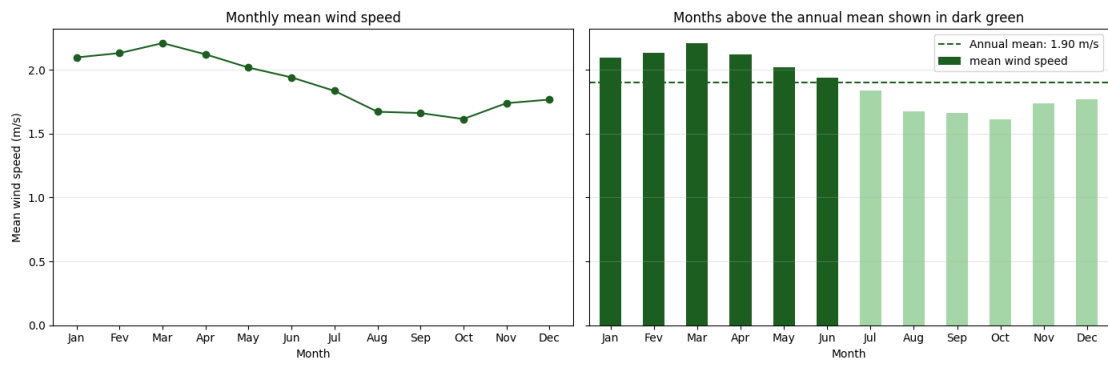
Canton GR — Wind seasonality, 2015–2025



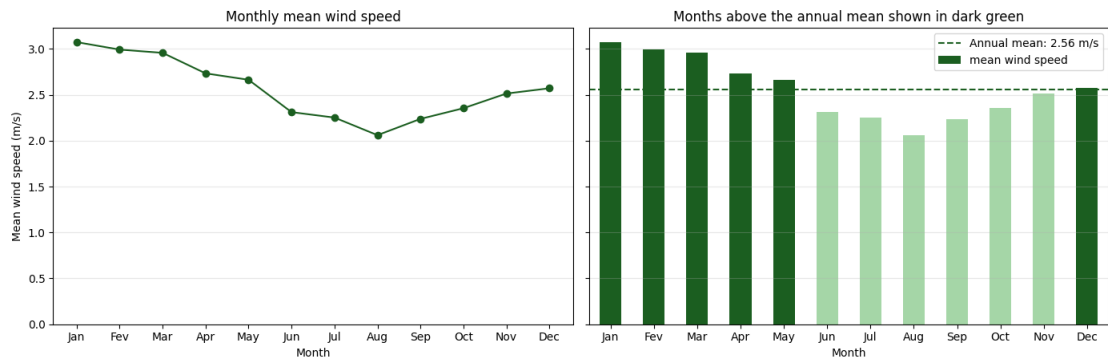
Canton JU — Wind seasonality, 2015–2025



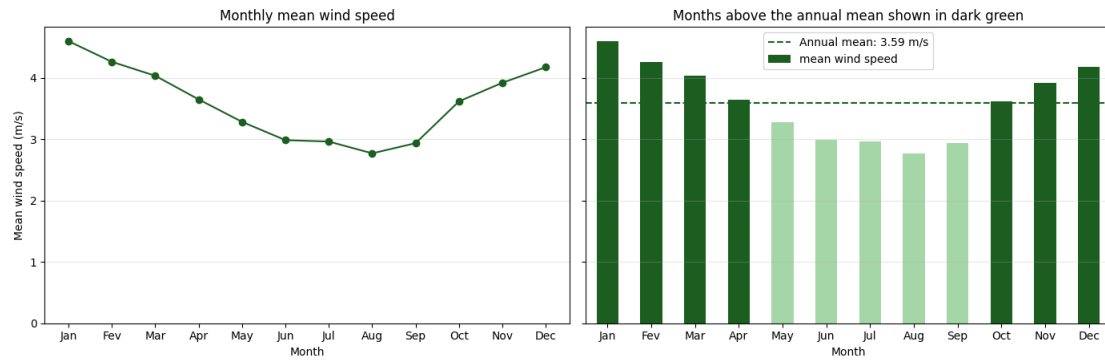
Canton LU — Wind seasonality, 2015–2025



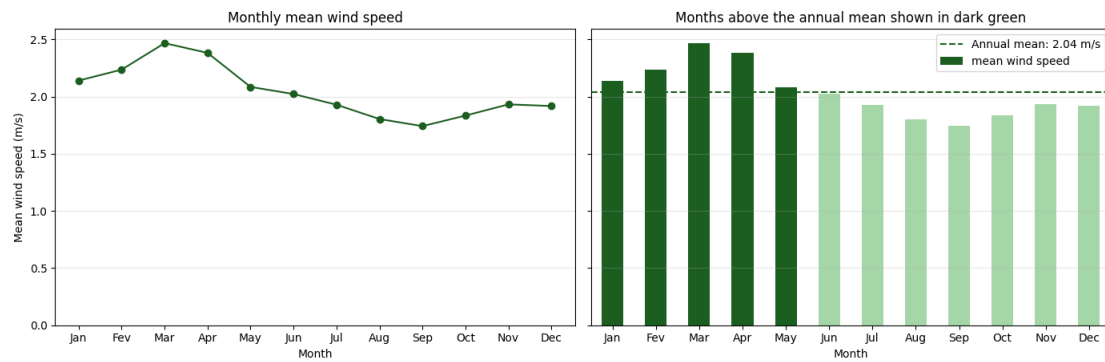
Canton NE — Wind seasonality, 2015–2025



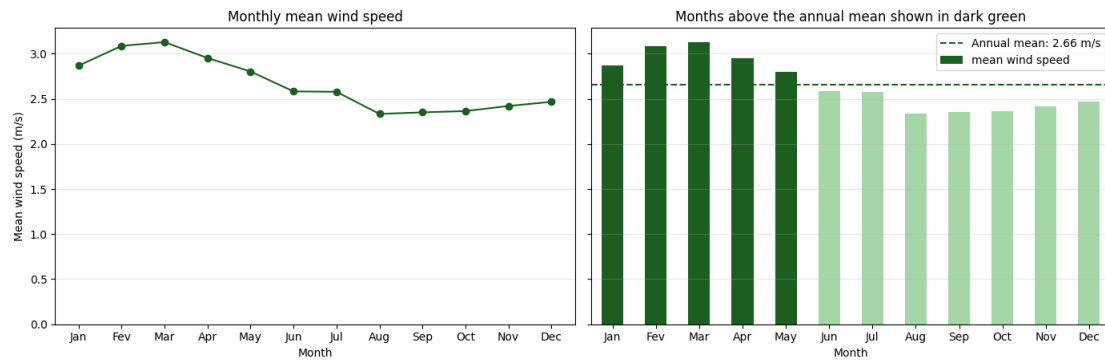
Canton OW — Wind seasonality, 2015–2025



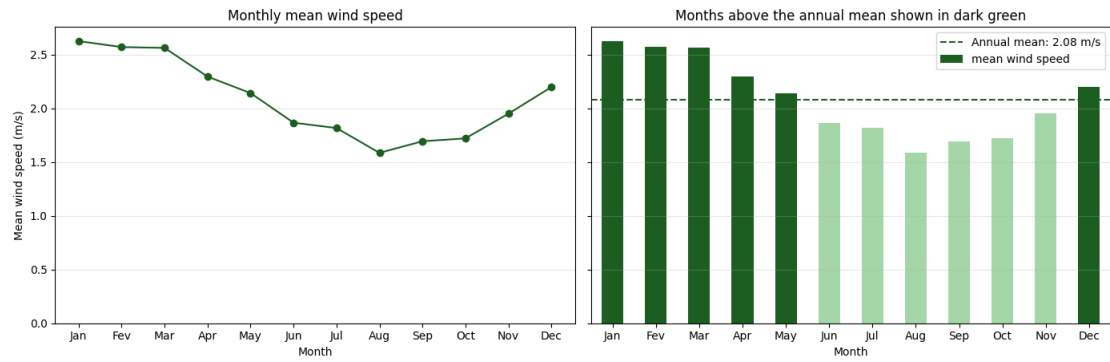
Canton SG — Wind seasonality, 2015–2025



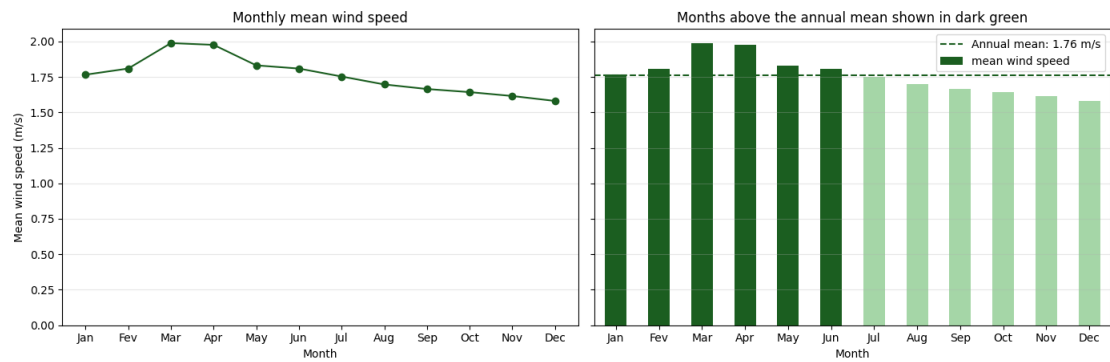
Canton SH — Wind seasonality, 2015–2025



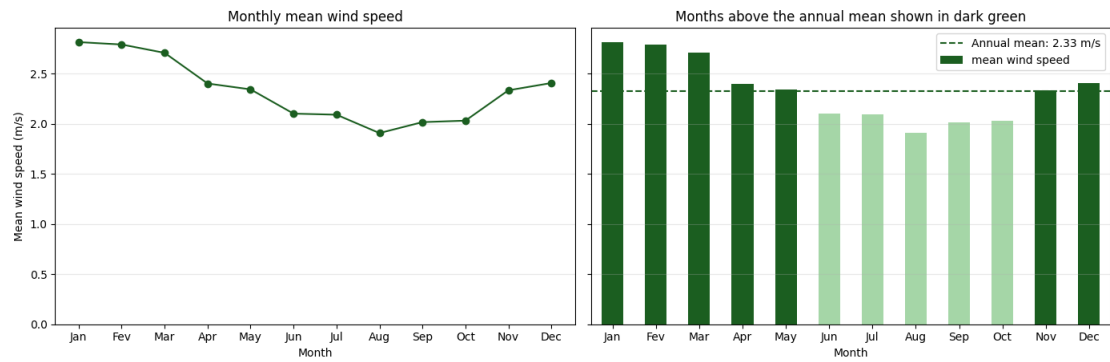
Canton SO — Wind seasonality, 2015–2025



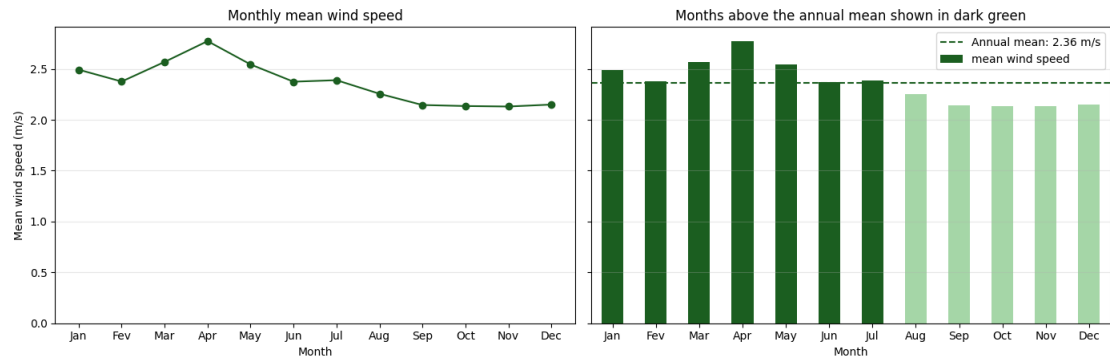
Canton SZ — Wind seasonality, 2015–2025



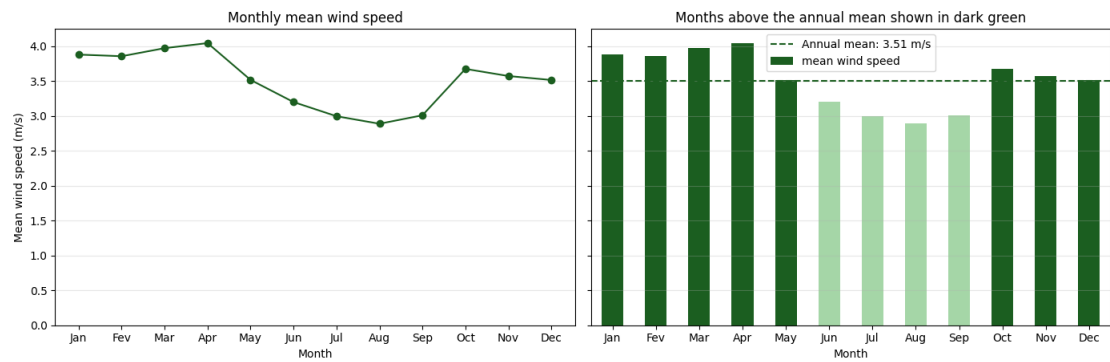
Canton TG — Wind seasonality, 2015–2025



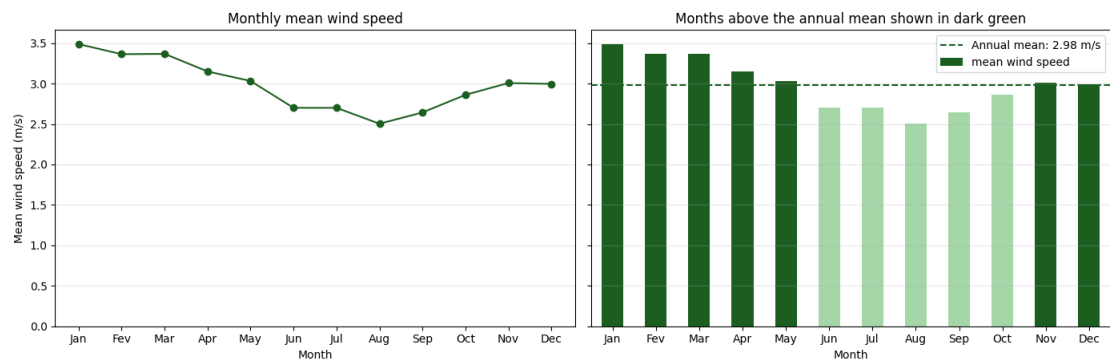
Canton TI — Wind seasonality, 2015–2025



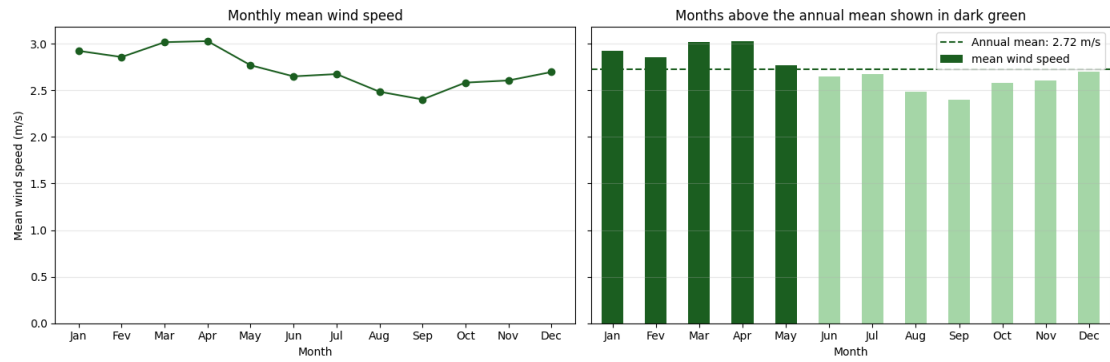
Canton UR — Wind seasonality, 2015–2025



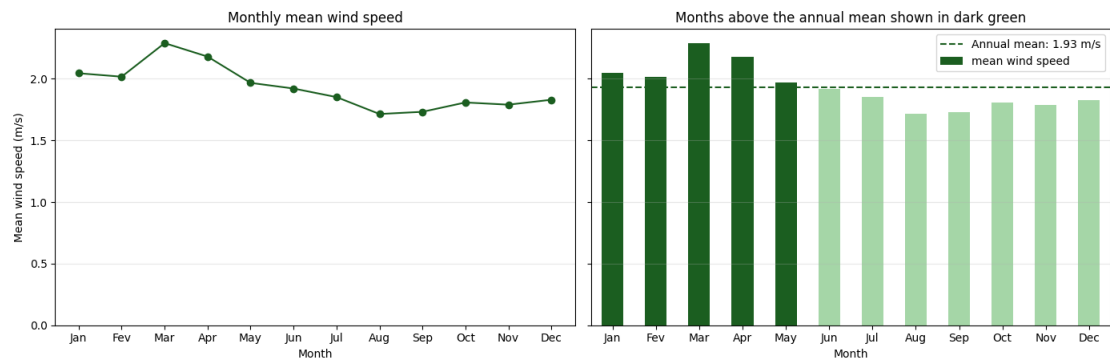
Canton VD — Wind seasonality, 2015–2025



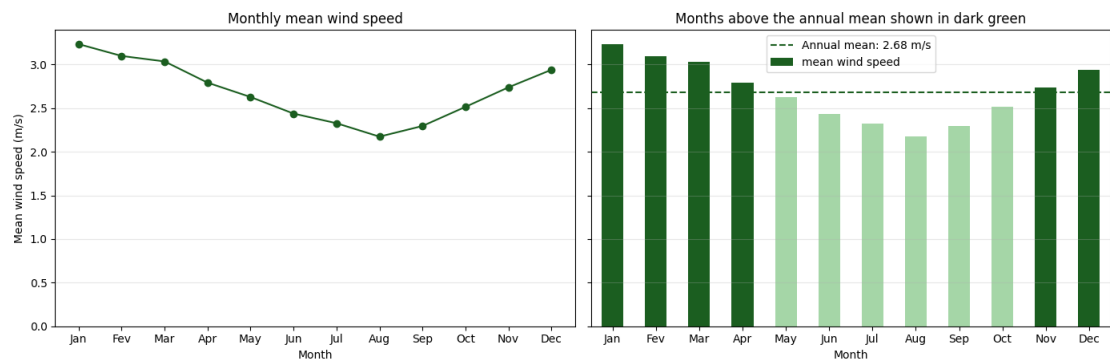
Canton VS — Wind seasonality, 2015–2025



Canton ZG — Wind seasonality, 2015–2025



Canton ZH — Wind seasonality, 2015–2025



Références

Marine Cauz, Nicolas Wyrsh, Lionel Perret, Christophe Ballif, and Alejandro Pena-Bello. Embracing wind power in the solar pv-dominated swiss landscape. *Energy Reports*, 2025. doi : 10.1016/j.egy.2025.02.059.